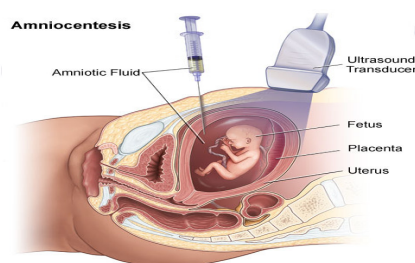


Guidelines for submission of amniotic fluid specimen for cytogenetic analysis



Collect in these single packed sterile Falcon tubes only.

Collection Details of Amniotic fluid / Chorionic Villi for Cytogenetic Studies

	Amniotic fluid	Chorionic Villi
Gestational Age for Collection	Can be done 15 – 20 weeks. Ideal time 16 – 18 weeks	Ideal time 11 weeks
Amount Needed	20-30 ml The first few mls of fluid are not used for cytogenetic analysis but may be used for AFP and /or ACHE	20mg of Chorionic villi in tissue culture medium
Quality	Sterile and Clear	Sterile
Collection / Transport Container	Sterile centrifuge tubes. Seal the tubes with parafilm sealer The tubes should be leak-proof.	1. Collection in the sterile medium provided. 2. Assess the quality and quantity of the villi
Specimen Labeling	All specimen containers must be clearly labeled with the patient's full name. Complete clinical history required along with gestational age and duly signed consent form	
Storage Conditions	Room temperature, protect from light. The specimen must be delivered immediately to the lab.	Room temperature. The specimen must be delivered immediately to the lab.
Transport Temperature	ROOM TEMPERATURE DO NOT FREEZE	
Transport Details	• Transport the sample immediately after collection . • Specimen should reach to the lab within 24 Hrs of collection to maintain viability of cells.. • Prior intimation to the centre is mandatory. • Consent form is mandatory. Strictly avoid collection of sample on Saturdays and on holidays.	

Suboptimal samples:

- Inadequate amount of specimen (<10-15ml of amniotic fluid/20-30 mg of chorionic villus sampling) may not permit adequate analysis.
- Exposure of the specimen to temperature extremes may kill cells and severely interfere with attempts to culture cells.
- If specimen reaches the lab after 24hrs, there is increased risk of result failure as contamination increases leading to death of living cells.
- Improper packaging may result in broken, leaky, and contaminated specimens during transport.
- Extended transport time.

Limitations

Cytogenetic studies on amniotic fluid/chorionic villus sampling (CVS) are considered more than 99% reliable for the detection of most fetal chromosome abnormalities. However, subtle or cryptic abnormalities involving microdeletions usually can be detected only with the use of targeted FISH testing.

A normal karyotype does not rule out the possibility of birth defects, such as those caused by submicroscopic cytogenetic abnormalities, molecular mutations, and environmental factors (ie, teratogen exposure). For these reasons, clinicians should inform their patients of the technical limitations of chromosome analysis before the procedure is performed, so that patients may make an informed decision about pursuing the procedure.